Quantitative Aptitude & Speed Math Formulas for Placements & CAT

High-yield mental calculation shortcuts, fractional percentage equivalences, profit & loss, time and work equations, and permutations/probability formulas.

LITHAVEN CAMPUS STUDY VAULT • OFFICIAL REVISION CAPSULE
DISCIPLINE: Aptitude & Mental Arithmetic | **TARGET:** Campus Placements, CAT, GATE General Aptitude & Banking

1. Percentage & Fractional Decimal Equivalences Table

Fraction	Percentage	Fraction	Percentage	Fraction	Percentage	Fraction	Percentage
1/2	50.00%	1/6	16.67%	1/10	10.00%	1/14	7.14%
1/3	33.33%	1/7	14.28%	1/11	9.09%	1/15	6.67%
1/4	25.00%	1/8	12.50%	1/12	8.33%	1/16	6.25%
1/5	20.00%	1/9	11.11%	1/13	7.69%	1/20	5.00%

2. High-Yield Commercial Math & Work Formulas

Topic Area	Core Shortcut Formula	Practical Calculation Example
Successive Percentages	Net % Change = a + b + (ab / 100)	+20% and +30% -> 20 + 30 + (600/100) = +56% net increase.
Profit & Loss / Dishonest Dealer	Gain % = [Error / (True Value - Error)] * 100	Dealer uses 900g instead of 1kg: [100 / (1000 - 100)] * 100 = 11.11% gain.
Time & Work Combined	Together Time = (A * B) / (A + B)	A does in 10 days, B in 15 days: (10 * 15) / (10 + 15) = 150 / 25 = 6 days.
Pipes & Cisterns	Net Rate = 1/Inlet_1 + 1/Inlet_2 - 1/Leak	Inlet fills in 6 hrs, leak empties in 8 hrs: 1/6 - 1/8 = 1/24 (fills in 24 hrs).
Relative Speed	Same Dir: |S1 - S2|	Opp Dir: S1 + S2	Trains of lengths L1, L2 crossing: Time = (L1 + L2) / Relative Speed.
Boats & Streams	Upstream U = B - S	Downstream D = B + S	Boat speed B = (D + U) / 2	Stream speed S = (D - U) / 2.

3. Speed Arithmetic & Combinatorics Mental Checklist

- **Squaring numbers ending in 5:** Take prefix n, multiply by n+1, append 25. Example: 85^2 = 8*(9)|25 = 7225.
- **Base 100 Multiplication:** 96 * 97. Deviations are -4 and -3. Left part: 96 - 3 = 93. Right part: (-4)*(-3) = 12 -> 9312.
- **Permutations vs Combinations:** Use nPr when order matters (seating, ranks). Use nCr when order is irrelevant (committees, card hands).
- **Circular Permutations:** Number of ways to arrange n distinct objects around a circle = (n - 1)!. For necklaces/garlands = (n - 1)! / 2.